

File size: This is the size of the file in bytes.

Ownership: The file owner and the group are determined by their user ID (UID) and group ID (GID).

Permissions: Read, write, and execute permissions for the file define access control.

Timestamps: Inodes record timestamps that provide modification (mtime), last access (atime), and inode change times (ctime).

Data block locations: Data block locations are pointers (addresses) to the physical locations on the disk where file data is stored.

Number of links: This refers to the number of hard links that point to the inode.

By efficiently storing metadata, inodes enable robust file access control mechanisms and facilitate auditing processes essential for maintaining system security and integrity.

Total number of inodes

Every system has numerous inodes, and it is important to be mindful of a few key numbers. Theoretically, the maximum number of inodes is equal to 2^{32} (approximately 4.3 billion). This is because of the potential range of the 32-bit unsigned integer data type, which is typically used for storing inode numbers. The number of

inodes in a filesystem is determined when it is created (formatted) based on factors such as partition size, desired block size, and filesystem type. Hence, a cap on the number of inodes is established when a filesystem is created. This implies that, even if there is free disk space, there is a possibility of running out of inodes if numerous small files are created.

Filesystems are typically configured with a default ratio of inodes to storage space (e.g., 1 inode per 16KB). The ratio is set when the filesystem is created. If you anticipate storing many very small files, you may choose a different ratio, such as one inode per 4KB, if the filesystem creation tools permit it.

Importance of inodes

- Inodes are essential for the Linux system to manage and access files efficiently. They enable quick retrieval of file metadata and content.
- Inodes manage file metadata by storing information, such as ownership, permissions, and timestamps.
- Inodes enable the system to locate all fragments of a file efficiently, even if these are scattered across different parts of the storage device.
- Multiple hard links can point to the same inode. This allows a file to have multiple names within the same filesystem.

Table 1 shows how you can interact with inodes using Linux commands.

```
student@GLB-CLAB-12:~$ ls -l -i
total 44
524721 drwxr-xr-x 3 student student 4096 Mar 10 12:16 Desktop
524725 drwxr-xr-x 3 student student 4096 Mar 26 15:26 Documents
524722 drwxr-xr-x 2 student student 4096 Mar 14 10:17 Downloads
524726 drwxr-xr-x 2 student student 4096 Jul 6 2024 Music
524727 drwxr-xr-x 3 student student 4096 Mar 5 15:33 Pictures
524724 drwxr-xr-x 2 student student 4096 Jul 6 2024 Public
535408 -rw-rw-r-- 1 student student 21 Mar 26 15:35 sample1
524393 drwx----- 7 student student 4096 Sep 26 15:22 snap
524723 drwxr-xr-x 2 student student 4096 Jul 6 2024 Templates
534708 drwxrwxr-x 2 student student 4096 Mar 5 15:32 test
524728 drwxr-xr-x 2 student student 4096 Jul 6 2024 Videos
student@GLB-CLAB-12:~$
```

Figure 1: Output of the `ls -l -i` command

Table 1: Commonly used inode commands

Command	Description	Usage example
<code>ls -l -i</code>	The location's files and directories are listed along with their inode numbers.	<code>ls -l -i</code>
<code>stat</code>	Displays detailed information about a directory or a file, including its inode number.	<code>stat filename</code>
<code>df -i</code>	Gives inode usage information for all the mounted filesystems, including the total number of inodes and used inodes.	<code>df -i</code>
<code>df -hi</code>	Displays the above information in human-readable format	<code>df -hi</code>
<code>du --inode</code>	Displays the number of inodes used by files and directories in the current directory.	<code>du --inode</code>
<code>find</code>	Searches for files by inode number, and can also execute commands on found files.	<code>find / -inum inode_number -exec rm -i {} \;</code>